

LESSON 9.5

EQUIVALENT NUMERICAL EXPRESSIONS



LESSON INTRODUCTION

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

LEARNING TARGETS

- I can determine if numerical expressions are equivalent.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

WORKING TOWARD

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

ESSENTIAL QUESTIONS

- What does it mean for numeric expressions to be equivalent?

LESSON NARRATIVE

In this lesson, students explore the concept of equivalence using numeric expressions. Students apply the order of operations with rational numbers to find the value of numeric expressions. In the guided component of the lesson, students investigate a variety of ways to rewrite equivalent expressions, including varying the placement of the negative and replacing decimals with their equivalent fraction representations (or vice versa). Ultimately, students are asked to simplify numeric expressions to determine if they are equivalent. The goal of these first lessons for MA.7.AR.1.2 is to review the concept of equivalence and properties of operations with known values before moving into algebraic expressions.

COMPONENT SUMMARY

9.5.1 WARM-UP (~5 MIN.)

Which One Doesn't Belong

9.5.3 GUIDED INSTRUCTION (10 MIN.)

Find the value of numerical expressions, understanding if the value is the same, then the expressions are equivalent

9.5.5 COLLABORATION (10 MIN.)

Create a numeric expression that is equivalent to a chosen value and verify their partner's expression is equivalent to the value

9.5.2 EXPLORATION (10 MIN.)

Explore equivalent numerical expressions

9.5.4 TRY IT (10 MIN.)

Evaluate numerical expressions and determine which are equivalent

9.5.6 WRAP-UP (~5 MIN.)

Determine if numeric expressions are equivalent

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Equivalent

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Consider creating reference sheets or posters that outline the order of operations and how to convert fractions to decimals (and vice versa).
- (Optional) Create a checklist with visuals to accompany 9.5.5 Collaboration.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not compute rational number operations fluently or accurately, hindering their ability to accurately determine if expressions are equivalent.
- Students may not follow the order of operations when performing the calculations in 9.5.2 Exploration, 9.5.3 Guided Instruction, and 9.5.4 Try It.
- Students may struggle creating an expression that simplifies to a given value in 9.5.5 Collaboration.

THINGS TO CONSIDER

- Encourage students who are not comfortable performing calculations with fractions to convert numbers to decimals, such as 0.5 instead of $\frac{1}{2}$, where appropriate and when this does not result in a repeating decimal.

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong?

9.5.1 Warm-Up provides an opportunity to encourage flexible thinking and multiple pathways by prompting students to choose at least two choices and providing rationale for why each does not belong.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MTR.MTR.4.1 Analyze Mathematical Thinking

9.5.5 Collaboration provides an opportunity for students to analyze the thinking of others and articulate their own thinking as they develop an understanding of how to create equivalent numeric expressions and confirm if expressions are equivalent.

DIFFERENTIATE INSTRUCTION

Transform 9.5.4 Try It into a Card Sort for students to physically sort and organize expressions based on their equivalent values. Consider your students when determining whether students are working independently or with a partner. If working with a partner, consider assigning specific roles for each student so that one student does not do all of the work. One student could be the scribe who writes the work and reasoning to be given to the teacher while one student could be the microphone. The scribe can only talk in the form of questions, and the microphone can only answer the questions asked. Strategically pairing a student who needs support as the scribe and a student fluent in fractions as the microphone provides entry points for both students in engaging with the 9.5.4 Try It.

9.5.1 WARM-UP: WHICH ONE DOESN'T BELONG?

TEACHER MOVES

Since the Warm-Up only involves integers, it provides an excellent opportunity to review the order of operations.

QUESTIONING

Challenge students to come up with a reason why each expression could be considered the one that does not belong.



9.5.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong? Justify your reasoning.

$(13 - 12 \div 2) + 3 \times 2$	$12 + (9 \div 3)^2$
$20 - 16 \div 8 + 5$	$38 - (2 + 4) + 20 - 3$

Answers will vary.



9.5.2 Exploration: Equivalent Numerical Expressions

1. Is $14 - \frac{3}{4}(12)$ equivalent to $\frac{47}{5} - (\frac{44}{5} \div 2)$? Explain how you determined your answer.

$$14 - \frac{3}{4}(12) = 14 - 9 = 5$$

$$\frac{47}{5} - (\frac{44}{5} \div 2) = \frac{94}{10} - \frac{44}{10} = \frac{50}{10} = 5$$

Yes, they are equivalent. I used the order of operations to find the value of expression and they both have the same value, 5.

2. Consider the expression $-\frac{3}{4}(12)$.

- a. What is the value of $-\frac{3}{4}(12)$?
-9

- b. Explain if $-\frac{3}{4}(12)$ can be rewritten as $0.75(-12)$?
Yes, this expression also has a value of -9.

- c. Does $\frac{-3(12)}{4}$ have the same value as the previous expressions?
Yes, this expression also has a value of -9.

- d. Create another expression equivalent to $-\frac{3}{4}(12)$. Share your answer with your partner.
Answers will vary. For example: $-\frac{15}{5}(3) = -9$ or $\frac{30}{10}(-3) = -9$ The value of the expression must be -9 to be equivalent.

The expressions in the previous problem all have the same value, but they are written differently.

3. What do you notice as the similarities and differences between $-\frac{3}{4}(12)$, $0.75(-12)$, and $\frac{-3(12)}{4}$?
Answers will vary.
The similarities include: all have one negative factor and all have three fourths multiplied by 12. They all have the same value.
Some differences include: two have three fourths expressed as a fraction and one has it expressed as a decimal. Some have the first factor negative and others have the second factor negative.

4. Explain which representation is easiest for you to evaluate using mental math.

Answers will vary.
For me, the third expression is the easiest to evaluate, because I can just evaluate the numerator and then divide by the denominator in my head.

9.5.2 EXPLORATION: EQUIVALENT NUMERICAL EXPRESSIONS

ENRICHMENT

If students get stuck with $(\frac{44}{5} \div 2)$, rewrite the expression as $(\frac{44}{5} \cdot \frac{1}{2})$ or prompt with questions like:

- How can division by 2 be rewritten using multiplication?
- What is 44 divided by 2? So, how many fifths is $\frac{44}{5}$ divided by 2?

TEACHER MOVES

Consider implementing a Notice and Wonder routine for question 3, positioning students to formalize their findings in the multiple representations of the expression and refine their planning and decision-making in future expressions.

9.5.3 GUIDED INSTRUCTION: EQUIVALENT NUMERICAL EXPRESSIONS

TEACHER MOVES

Before question 1, implement a Turn and Talk routine and have students discuss what it means for numeric expressions to be equivalent. Include a required step in which students discuss which numeric expressions from the exploration are equivalent (all of them).

DIFFERENTIATE INSTRUCTION

Encourage students to use fractions or decimals in their responses to continue spiraling fluency with fractions and reducing anxiety.



9.5.3 Guided Instruction: Equivalent Numerical Expressions



Numerical expressions are equivalent if their values, when simplified, are the same.

Algebraic expressions are equivalent if they have the same value no matter what number is substituted for each of the variables.

- Are the following expressions equivalent? Show your work or explain your thinking.

$$\frac{9}{4} + \left(3.5 - 7\frac{1}{2}\right) \text{ and } \left(-\frac{1}{2} \cdot \frac{3}{4} + 6.5\right) - 7\frac{7}{8}$$

$$\frac{9}{4} + \left(3.5 - 7\frac{1}{2}\right) = \frac{9}{4} - 4 = \frac{9}{4} - \frac{16}{4} = -\frac{7}{4} = -1\frac{3}{4}$$

$$\left(-\frac{1}{2} \cdot \frac{3}{4} + 6.5\right) - 7\frac{7}{8} = -\frac{3}{8} + \frac{13}{2} - \frac{63}{8} = -\frac{3}{8} + \frac{52}{8} - \frac{63}{8} = -\frac{14}{8} = -1\frac{3}{4}$$

Yes, they are equivalent because both expressions have the same value.

- Work with your partner to create a third expression that is also equivalent to $\frac{9}{4} + \left(3.5 - 7\frac{1}{2}\right)$.

Answers will vary. For example: $2 \cdot 4.25 - 10.25 = -1.75$. Any expression with a value of $-1\frac{3}{4}$ or -1.75 is an equivalent expression.



9.5.4 Try It: Equivalent Numerical Expressions

1. Evaluate expressions A-L.

A $-\frac{1}{3}(-24) + 3$ $= 8 + 3$ $= 11$	B $(-2.25 \times 4) \cdot 2$ $= -9 \cdot 2$ $= -18$	C $-\frac{15}{15} + 4\frac{1}{3}$ $= 3\frac{1}{3}$
D $3(2.5 - 6) + 7\frac{1}{2}$ $= 3(-3.5) + 7.5$ $= -10.5 + 7.5 = -3$	E $4\left(-\frac{1}{4}\right)(1)(-1^2)$ $= -1(1)(-1)$ $= 1$	F $3 + \frac{-24}{-3}$ $= 8 + 3$ $= 11$
G $\frac{5(-4\frac{1}{4} + 6.25)}{3}$ $= \frac{10}{3}$ or $3\frac{1}{3}$	H $-6\frac{2}{3} \cdot 2.4 \div (-4)$ $= \frac{-20}{3} \cdot \frac{12}{5}$ $= -4$ $= \frac{-16}{-4} = 4$	I $-7.5 + \left(6\frac{1}{6} \cdot 3\right)$ $= -7.5 + \left(\frac{37}{6} \cdot \frac{3}{1}\right)$ $= -7.5 + 18.5$ $= 11$
J $\left(-4.2 + 9\frac{4}{5}\right) + \left(1 + 4\frac{4}{10}\right)$ $= \left(-\frac{42}{10} + \frac{98}{10}\right) + \left(\frac{10}{10} + \frac{44}{10}\right)$ $= \frac{56}{10} + \frac{54}{10} = 11$	K $-\left(\frac{-18}{-5}\right) - 0.4$ $= -\frac{18}{5} - \frac{2}{5}$ $= -\frac{20}{5}$ $= -4$	L $\left -3\frac{1}{8} - 0.875\right $ $= -3.125 - 0.875 $ $= -4 $ $= 4$

2. Sort the expressions into groups based on equivalence. Record the letters of the equivalent expressions below.

Equivalent Expressions
<i>A, F, I, and J are equivalent. They all have a value of 11.</i>
<i>H and L are equivalent. They both have a value of 4.</i>
<i>C and G are equivalent. They both have a value of $3\frac{1}{3}$.</i>

9.5.4 TRY IT: EQUIVALENT NUMERICAL EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Consider providing the expressions as cut-outs to transform this activity into an interactive Card Sort.

QUESTIONING

Advanced learners may be able to perform calculations using mental math. Consider challenging these students to solve using mental math and describe their strategies.

For example, students may show evidence of mastery of MA.K12.MTR.3.1 in card J by exhibiting efficiency in mentally converting $9\frac{4}{5}$ to $9\frac{8}{10}$ because they explain that they recognize -4.2 as $(-4\frac{2}{10})$ as well, identifying the common denominator as tenths mentally.

9.5.5 COLLABORATION: GENERATING EQUIVALENT EXPRESSIONS

TEACHER MOVES

Students have an opportunity to analyze the thinking of others and articulate their own thinking as they develop an understanding of how to create equivalent numeric expressions and confirm if expressions are equivalent.

Consider providing students with question stems to aid in their discussion. For example:

- How did you create your numeric expression? How do you know it is equivalent?



9.5.5 Collaboration: Generating Equivalent Expressions

1. Work with a partner to choose a numerical value.

Individually, create a unique expression using rational numbers that equals the numerical value chosen when evaluated. Include a **minimum** of each of the following:

- four operations
- one grouping symbol
- two rational numbers written as fractions

When you are finished creating your expression, trade and check each other's numeric expressions.

Answers will vary. One example below.

Numerical Value <u>10</u>	
Your Expression	Your Partner's Expression
$\left(7\frac{1}{4} - 2\frac{1}{4} + 20\right) \times 2 \div 5$	$\left(12\frac{1}{2} - 2\frac{1}{2} + 5\right) \times 4 \div 6$

2. Are they equivalent? If not, correct any errors.

Yes, are equivalent. They both have a value of 10.



9.5.6 Wrap-Up: Cool Down

1. Are the following expressions equivalent? Show your work or explain your reasoning.

$$\left(-7.6 + 2\frac{1}{2}\right)(-6) \text{ and } (-8)\left(-3\frac{1}{10}\right) + \frac{2}{3}(8.6)$$

No, they are not equivalent expressions, because they do not have the same value.

$$\begin{aligned} &\left(-7.6 + 2\frac{1}{2}\right)(-6) \\ &= -7.6 + 2.5(-6) \\ &= (-5.1)(-6) \\ &= 30.6 \end{aligned}$$

$$\begin{aligned} &(-8)\left(-3\frac{1}{10}\right) + \frac{2}{3}(8.6) \\ &= (-8)\left(-\frac{31}{10}\right) + \frac{2}{3}\left(\frac{43}{5}\right) \\ &= \frac{124}{5} + \frac{86}{15} \\ &= \frac{372}{15} + \frac{86}{15} \\ &= \frac{458}{15} \\ &= 30.5\bar{3} \end{aligned}$$

9.5.6 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

Consider giving students an alternative option to provide student choice.

- “Write your own three numeric expressions. Two of the expressions must be equivalent, and all three expressions must have at least one fraction or decimal.”